ModelIt!

Student Activity Pack

Why Does a Glow Stick Stop Glowing?

Modeling Chemical Kinetics and Energy Transformation in Chemiluminescence

Grade 10L2

NGSS HS-PS1-4, HS-PS1-6

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think makes a glow stick glow — and why does it eventually stop?

2 Why does putting a glow stick in hot water make it glow brighter? What happens when you put it in the freezer?

3 Draw a diagram showing what you think is happening at the molecular level inside a glowing glow stick.

4 What is the relationship between temperature and chemical reaction speed?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Chemical Concentration | Temperature | Reaction Rate | Fluorescent Dye Activation | Light Output | Energy Dissipation | Product Buildup

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Room Temperature

Standard conditions, 25 degrees C

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Hot Water Bath

Elevated temperature, 60 degrees C

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Freezer Storage

Low temperature, -10 degrees C

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Catalyst Presence — A substance that lowers the activation energy of the reaction without being consumed, allowing the reaction to proceed faster at any given temperature — some glow stick formulations include catalysts to enhance initial brightness

Dye Quantum Yield — The efficiency of the fluorescent dye in converting chemical excitation energy into visible light photons versus heat — higher quantum yield means more light per unit of chemical reaction

Solvent Viscosity — The resistance of the liquid medium to molecular movement, which affects how quickly reactant molecules can diffuse and collide — lower viscosity at higher temperatures contributes to faster reaction rates

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does a glow stick glow brighter when you heat it up, dimmer when you cool it — and why does it eventually stop glowing entirely? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the relationship between activation energy and temperature in determining reaction rate?**

2. **Why is the trade-off between brightness and duration a fundamental constraint rather than an engineering limitation that could be overcome?**

3. **What does your model predict about total light output at different temperatures — does a hot glow stick produce more total light than a cold one?**

4. **How does Product Buildup create a negative feedback loop that gradually extinguishes the glow stick?**

5. **What are the limitations of modeling chemiluminescence with only seven components when real glow stick chemistry involves dozens of intermediate reactions?**

8 STEM Challenge: Design an Optimized Emergency Light Stick

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YOUR ENGINEERING MISSION

Design a chemiluminescent light source optimized for a specific emergency scenario — maximizing either brightness, duration, or total visibility based on mission requirements.

THE SCENARIO

A search-and-rescue equipment company needs light sticks for three different scenarios: maritime distress signals (maximum brightness for 30 minutes), camping emergency lighting (moderate brightness for 12+ hours), and underwater cave markers (dim but visible for 48+ hours). Your team must design the chemical formulation and temperature management strategy for each.

Engineering Design Thinking

1. WHAT REACTANT CONCENTRATIONS WOULD YOU USE FOR THE 30-MINUTE VERSUS 48-HOUR LIGHT STICK?
2. HOW COULD YOU INCORPORATE A THERMAL MANAGEMENT SYSTEM TO CONTROL GLOW RATE IN FIELD CONDITIONS?
3. WHAT PACKAGING MODIFICATIONS WOULD HELP MAINTAIN OPTIMAL TEMPERATURE IN MARITIME VERSUS CAVE ENVIRONMENTS?
4. HOW WOULD YOU TEST AND VALIDATE YOUR DESIGNS USING MODEL PREDICTIONS BEFORE MANUFACTURING PROTOTYPES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Chemical Engineers design chemical processes and products for manufacturing, including specialty chemicals, materials, and consumer products. They work in industries from pharmaceuticals to energy, earning $75,000–$140,000/year. Photochemists who study light-producing chemical reactions earn $65,000–$130,000/year.